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Hackenbush Bamboo

CPU TIME LIMIT
1 second

MEMORY LIMIT
1024 MB

2.0

DIFFICULTY
Easy

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SOURCE & LICENSE
 Atli Fannar Franklin
E-402-STFO 2024 Module H

Sample input 1

reach the ground are discarded.

The value of a Hackenbush game with only green edges is simply the Grundy number of that game.

You are given a position in Hackenbush bamboo. This is a position in the game Hackenbush where all edges are green (can be chopped by either player) and all edges are on straight paths from the ground, which do not split.

You need to determine the value of the game. As the game only has green edges this will always be an integral number.

Input

The first line contains a single integer $0 \leq n \leq 10^3$, the number of bamboo stalks in the game. The next line contains n integers $1 \leq h_i \leq 10^9$, the height of the i -th stalk.

Output

If the value of the game is $*k$ then print $*k$. The value of the game is always of this form.

Sample Input 1

```
5
2 3 2 1 4
```

Sample Output 1

```
* 6
```

Sample Input 2

```
5
3 3 1 2 3
```

Sample Output 2

```
* 0
```

